HOW DOES A CATALYST WORK?

Not all collisions result in a reaction happening. Collisions have to involve at least a certain minimum amount of energy, called the *activation energy* (page 230).



▲ Figure 20.24 Traffic passes easily through a road tunnel under a mountain.

If a reaction is slow, it means that very few collisions have this amount of energy; when most collisions happen, the particles simply bounce off each other.

Catalysts work by providing an alternative route for the reaction, involving a lower activation energy.

If the activation energy is lower, many more collisions are likely to be successful. The reaction happens faster because many more particles *have energy greater than or equal to the activation energy for the alternative route*. The energy of the particles has not changed but it has been made easier for them to react.

You can illustrate this with a simple everyday example. Imagine you have a mountain between two valleys. Only a few very energetic people will climb over the mountain from one valley to the next. Now imagine building a road tunnel through the mountain. Lots of people will be able to travel easily from one valley to the next.

Be careful how you phrase the statement explaining how a catalyst works. You should say 'Catalysts provide an alternative route with a lower activation energy'.

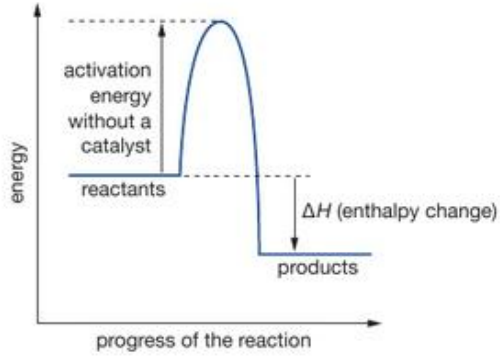
They *do not* 'lower the activation energy' any more than building a tunnel lowers the mountain. The original route is still there, and if particles collide with enough energy they will still use it, just as very energetic people will still choose to climb over the top of the mountain.

CHEMISTRY ONLY

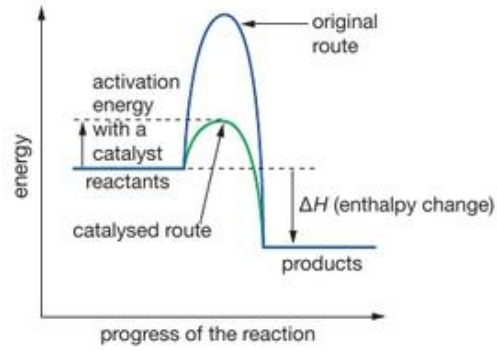
You can show the effect of a catalyst on a reaction profile diagram (Figure 20.25 and Figure 20.26). Adding a catalyst gives the reaction an alternative way to happen that involves a lower activation energy.

KEY POINT

The reaction here is an exothermic reaction: the products have lower energy than the reactants.



▲ Figure 20.25 A reaction profile diagram



▲ Figure 20.26 Activation energy with a catalyst

KEY POINT

Notice in Figure 20.26 that the difference in energy between the reactants and the products remains the same with or without a catalyst.

END OF CHEMISTRY ONLY

CATALYSTS IN INDUSTRY

Catalysts are especially important in industrial reactions because they help substances to react quickly at lower temperatures and pressures than would otherwise be needed. This saves money.

CHAPTER QUESTIONS

SKILLS CREATIVITY 5

1 A student carried out an experiment to investigate the rate of a reaction between an excess of dolomite (magnesium carbonate) and 50 cm^3 of dilute hydrochloric acid. The dolomite was in small pieces. The reaction is:

$$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$$

He measured the volume of carbon dioxide given off at regular intervals, with the results shown in the table below.

Time/s	0	30	60	90	120	150	180	210	240	270	300	330	360
Volume/cm^3	0	27	45	59	70	78	85	90	94	97	99	100	100

a Draw a diagram of the apparatus you would use for this experiment and label it.

SKILLS INTERPRETATION

b Plot these results on a piece of graph paper, with time on the *x*-axis and volume of gas on the *y*-axis.

SKILLS ANALYSIS, REASONING 6

c At what time is the gas being given off most quickly? Explain why the reaction is fastest at that time.

SKILLS PROBLEM SOLVING 5

d Use your graph to find out how long it took to produce 50 cm^3 of gas.

6

e Use your graph to calculate the average rate of this reaction in the first 80 seconds.

SKILLS REASONING

f In each of the following questions, state and explain what would happen to the initial rate of the reaction and to the total volume of gas given off if various changes were made to the experiment.

i The mass of dolomite and the volume and concentration of acid were kept constant, but the dolomite was in one big lump instead of small pieces.

ii The mass of dolomite was unchanged and it was still in small pieces. 50 cm^3 of hydrochloric acid was used, which had half the original concentration.

SKILLS REASONING

6

iii The dolomite was unchanged again. This time 25 cm^3 of the original acid was used instead of 50 cm^3.

iv The acid was heated to 40 °C before the dolomite was added to it.

7

2 The effect of concentration and temperature on the rate of a reaction can be explored using the reaction between magnesium ribbon and dilute sulfuric acid:

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

A student dropped a 2 cm length of magnesium ribbon into 25 cm^3 of dilute sulfuric acid (a large excess of acid) in a boiling tube. She stirred the contents of the tube continuously and timed how long it took for the magnesium to disappear.

a What would you expect to happen to the time taken for the reaction if the student repeated the experiment using the same length of magnesium with a mixture of 20 cm^3 of acid and 5 cm^3 of water? Explain your answer in relation to the collision theory.

b What would you expect to happen to the time taken for the reaction if she repeated the experiment using the original quantities of magnesium and acid, but first heated the acid to 50 °C? Explain your answer in relation to the collision theory.

c Give a reason why is it important to keep the reaction mixture stirred continuously.

CHEMISTRY ONLY

SKILLS INTERPRETATION

10

d The reaction between magnesium ribbon and dilute sulfuric acid is exothermic. Draw a reaction profile diagram for this reaction, label the axes clearly and show both the activation energy and enthalpy change on your diagram.

END OF CHEMISTRY ONLY

3 Catalysts speed up reactions, but can be recovered chemically unchanged at the end of the reaction.

SKILLS REASONING

a Explain how a catalyst works to increase the rate of reactions.

SKILLS DECISION MAKING

b Describe how you would find out whether iron(III) oxide was a catalyst for the decomposition of hydrogen peroxide solution. Note: you need to show not only that it speeds the reaction up, but that it is chemically unchanged at the end.

CHEMISTRY ONLY

c Draw a clearly labelled reaction profile diagram showing the effect of using a catalyst in a reaction.

END OF CHEMISTRY ONLY